Meeting 3

This is the summary of the first meeting held with representative from institute B. The table summarizes the purposes, main topics discussed, and outputs of the meetings.

Third meeting	
Stakeholders	Institute B, representative B1 and B2.
Topics of discussion	We initiated discussions regarding defining the domain of reference of the ClimOO Ontology. We engaged with two stakeholders regarding perspectives on sustainability and related frameworks of reference, agri-food systems of interest and related boundaries, primary features to address with regard to plastic, plastic pollution, and climate change, and possible data to interoperate. Table 1 provides a summary of the outcomes of those discussions. Stakeholders provide us with a detailed list of replies to the questions we furnished them as guidelines for the discussion. We met with only one of the stakeholders due to the stakeholders' schedule commitments. Replies, however, contained perspectives of both involved stakeholders.
Stakeholder perspective on sustainability	Sustainability in terms of "our common future"; the Brundtland Report. Sustainability in terms of the "three pillars" view: environmental, social, and economic dimensions.
Framework of reference on sustainability, plastics, and climate change	The South Tyrolean apple sector's sustainable development strategy: The ABCs of sustainability. https://www.sustainapple.it/en/ Sustainapple - VIP Coop. soc. Agricola. https://www.vip.coop/en/business/sustainability/sustainapple/159-0.html L'Agenda 2030 del Consorzio Vini Alto Adige. Centro di competenza per il vino dell'Alto Adige. https://www.vinialtoadige.com/it/home/1-0.html Bioland Südtirol guidelines https://www.bioinsuedtirol.it/ AGRIOS guidelines. Gruppo di lavoro per la frutticoltura integrata. https://www.agrios.it/it/ Demeter guidelines. Demeter. https://www.demeter.it/ EU Commission Press Release ip_23_4581. https://ec.europa.eu/commission/presscorner/detail/en/ip_23_4581

	EU Common Agriculture Policy (CAP). https://agriculture.ec.europa.eu/common-agricultural-policy/cap-overview/cap-glance_en EU Corporate Sustainability Reporting Directive (CSRD). https://finance.ec.europa.eu/capital-markets-union-and-financial-markets/company-reporting-and-auditing/company-reporting/corporate-sustainability-reporting_en EU organic farming policy. https://agriculture.ec.europa.eu/farming/organic-farming/organics-glance_en EU EN 13432 bioplastic policy. https://www.en-standard.eu/bs-en-13432-2000-packaging.-requirements-for-packaging-recoverable-through-composting-and-biodegradation.-test-scheme-and-evaluation-criteria-for-the-final-acceptance-of-packaging/ EU Waste Codes: EU Waste Directives https://environment.ec.europa.eu/topics/waste-and-recycling/implementation-waste-framework-directive_en European Wastes Catalogue (EWC) https://www.eea.europa.eu/help/glossary/eea-glossary/european-waste-catalogue-1
Agriculture system of reference	Agricultural system in the South Tyrol region. Main system scales of analysis: spatial, social, and economic scales. Main levels of analysis: farm, region, national, and international levels. Main system components: farms, fields, inputs (seeds, fertilizers, pesticides), outputs (harvested products). Main system actors: Farms/businesses (especially family and small businesses; average ~3 ha in fruit growing), cooperatives, food retailers and distributors, and food processors. Additional points: possible frictions among system components due to living within the same space.
Relevant system stakeholders	Farmers, consultancy services, cooperatives, farmers' associations, political decision-makers (regional government), researchers, marketers, and the local and regional communities. Some cited examples:

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- The South Tyrolean Extension Service for Fruit and Wine Growing https://www3.beratungsring.org/it/home
- Consortium of South Tyrolean Fruit Growers' Cooperatives (VOG) https://www.vog.it/en/home.html

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<tr><td>Additional take-home messages regarding sustainability in agriculture</td><td>

To achieve:

- Judicious and efficient management of soil, water, and environment to ensure farm survival and production, and resource conservation (No resource exploitation).
- Avoiding unnecessary environmental contamination.
- Promoting biodiversity.
- Protecting and creating habitats.
- Forward-looking strategies, especially circular economy strategies.
- Adaptability to changes, especially changes in price pressure and regulatory requirements.
- Managing land prices and production costs.
- Ensuring the economic survival of businesses in small geographical areas.
- Incentivizing marketing support, such as direct marketing, farm sales, and cooperation with tourism.
- Guaranteeing fair working conditions and salary.
- Supplying local communities with local food.
- Safeguarding farm succession and farm internal social structure (e.g, caring for older generations).

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<tr><td>Primary features to address concerning plastic and plastic pollution in agriculture</td><td>

Plastics: synthetic or semi-synthetic materials made from polymers.

Main characteristics to refer to (depending on the type of plastic material considered): (not) degradable, (not) biobased, may fragment, durable, multi-functional, lightweight, low-cost, formable, and stable.

They have different rates of fragmentation and microplastic formation behaviors.

Attention to possible misunderstandings of what plastic degradability, compostability, and persistence in the environment are: according to EU EN 13432, PLA is fully degradable and mineralizes in six months under industrial composting conditions. Under natural conditions (open fields and home composts), degradation runs more slowly, and plastic polymers may take several years to degrade (high temperature and adequate microbial conditions are required).

Additives:

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	Leaching into the soil or water systems. Affecting plant growth, soil microbiota, and food safety. Interacting with other pollutants. Most relevant: Plasticizers (phthalates), UV stabilizers, and colorants (in mulching films, irrigation hoses, or fruit bags). Main polymers composing plastic materials used in agriculture: PE, PVC, PLA, PA, and PP. They have different rates of degradation. They have different chemical binding capacities for environmental contaminants. In agriculture, plastics are used differently depending on culture, crops, cultivation practices, and legally specified sectors of application. ● To protect crops (e.g., hail nets, insect nets). ● To manage soil and microclimate (e.g., mulching films, protective covers), (mulching films are not widely used in South Tyrol). ● To irrigate (e.g., drip lines). ● To support plants and bind materials (e.g., ties, clips, nets). ● Manage the release of cultivation inputs (pellets, granules, capsules, compost). ● Packaging and transport (e.g., crates, containers, fertilizer, and pesticide packaging). Short-term single-use plastic-based materials: largely used, not collected, and few recycling or reusage options. Long-term use plastic-based materials: widely used, and persistent in the system for several years. Plastics in South Tyrolean apple cultivation: • Plastic materials are used primarily for binding. • Limited use of plastic films in apple orchards (exception: fruit coloring films, but not standardized usage). • Limited use of compost. • Packaging materials: out of our system of interest. Microplastics: small synthetic plastic particles, under 5 mm, that are organic, insoluble, and do not degrade (U Commission Press Release ip_23_4581).

	Main characteristics to refer to: size, shape, color, and aggregates. Plastic and microplastic pollution: Plastic persistence in soils, breaking down into microplastics, and accumulation over time affects soil structure and porosity, water retention, aeration, and microorganisms. There may also be impacts on humus formation, water-nutrient availability, and contamination of groundwaters and surface waters. Plastic additives released in the soil may affect soil chemical and biological activities. Question: Which level of plastic accumulation corresponds to measurable effects? Efforts on plastic usage. Contributions to climate change: • Plastic production and disposal contribute to climate change, especially in terms of greenhouse gas emissions, energy consumption, and resource depletion. Usage of plastic during agricultural practices contributes to climate change less than these other activities. • Accumulation of microplastics in the soil may affect soil structure, soil fertility, and present consequences on CO_2 storage (reducing CO_2 binding in the soil). • Plastics are one of the main drivers of CO_2 emissions along the supply chains, from production to disposal. Recommendation of focusing on: Types of plastic materials and related properties. Accounting for differences between plastic materials and microplastics. Including details on microplastic adsorption/absorption of pollutants only if it becomes relevant. Including a general representation of additives by functional group. Further details are only for high-risk and toxic compounds.
Primary features to address concerning climate change in agriculture	Main levels of analysis: climatic conditions, consequences on agricultural systems, and physiological responses of plants. Consequences in agriculture:

	• Changes in the average temperature increase the risk of frost damage (late frost, early budburst or flowering), emergence of invasive pests, outbreak of novel diseases, and altered pest and disease pressures (no native natural enemies). • Heat damage and stress, and drought stress (depending on year and location). • Decrease in the quality of fruits, e.g., sunburn, flesh blemishes, reduced storability, physiological disorders, altered aroma development, and color formation. • An increase in production costs and the occurrence of extreme events affect agricultural production. • Generally, changing production conditions leads to an increase in uncertainties in yields, quality, and management strategies. Contributions to plastic pollution: • An increase in temperature, UV radiation intensity, and the occurrence of storms and floods leads to an increase in plastic debris erosion and fragmentation.

Primary features to address concerning plastic pollution and climate change in relation to the sustainability of agricultural systems.	Main sustainability issues concerning plastic pollution and climate change in agriculture:
	• Greenhouse gas emissions along the supply chains.
	• Extreme weather events.
	• Contamination of soil and water.
	    ○ Soil and water quality.
	    ○ Soil structure.
	    ○ Water retention capacity.
	    ○ Organic matter.
	• Resource consumption.
	• No circularity in the plastic life cycle.
	    ○ Environmental impacts of short-term single-use plastics.
	    ○ Few alternatives, generally too expensive.
	To achieve:
	• Promoting reusable, biodegradable, or recyclable plastics against single-use products.
	• Reducing plastic usage and implementing viable and sustainable alternatives. It is unclear which threshold of usage reduction should be reached.
	• Establishing plastic circular systems while keeping costs manageable.
	• Avoiding plastic accumulation in the agricultural systems.
	• Reducing greenhouse gas emissions.
	• Reducing the usage of raw materials.
	• Promoting the usage of energy from renewable sources.

	• Enhancing the usage of resource-saving cultivation and climate-smart methods, for instance, tailored fertilization and plant protection. • Fostering the cultivation of climate-change resilient varieties of plants.
Data of reference	To further discuss with other stakeholder institutes.
Relevant projects	See links cited above.